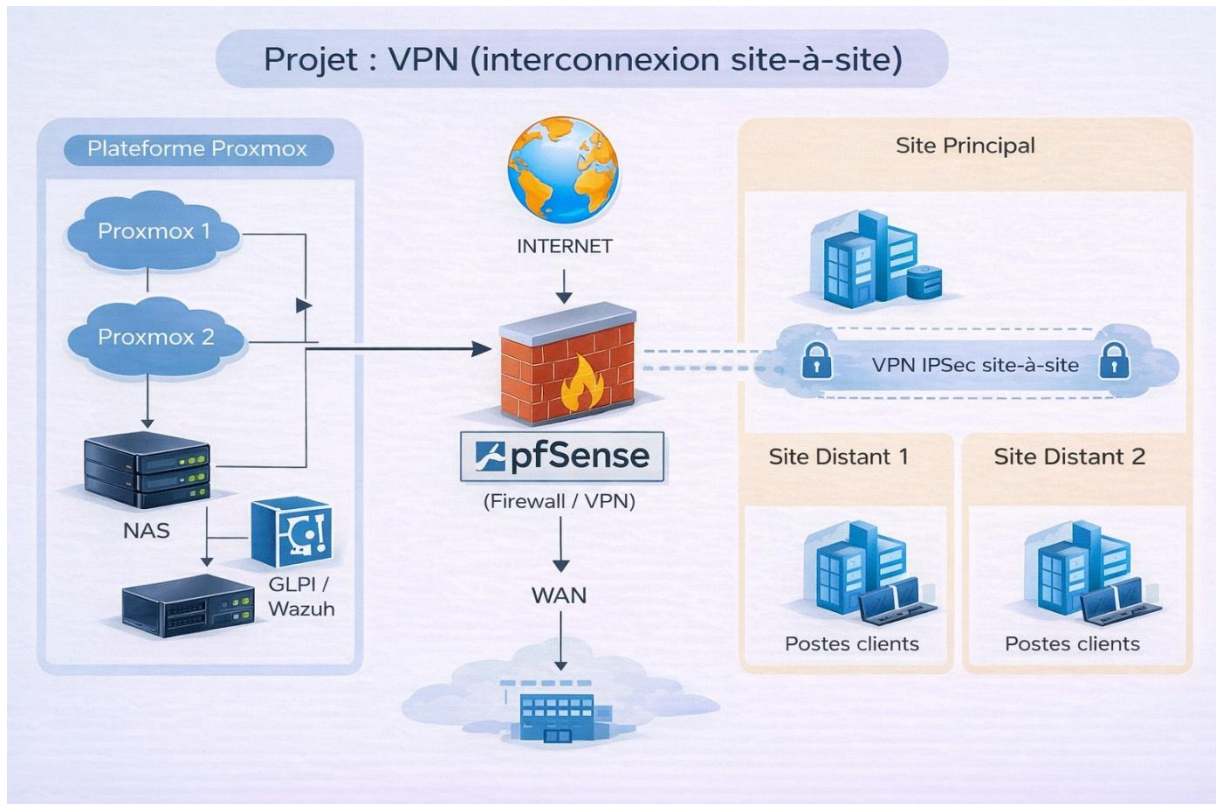


PLAN DU PROJET : VPN SITE TO SITE :)

ÉTAPE 1 — PRÉSENTATION DU BESOIN :

L'objectif est de relier deux réseaux distants de manière sécurisée à l'aide d'un VPN site-to-site basé sur le protocole IPsec.



ÉTAPE 2 — ARCHITECTURE RÉSEAU

Site A : PF101

- WAN : 192.168.101.1 (~~Site Simu : 192.168.50.1~~)
- LAN : 192.168.2.1 (~~site simu : 192.168.101.2.1~~)
- Réseau : 192.168.2.0/24

```
Press ENTER to continue.

QEMU Guest - Netgate Device ID: 71147942554b6699a1fd

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on PF101 ***

WAN (wan) -> vtnet0 -> v4: 192.168.101.1/24
LAN (lan) -> vtnet1 -> v4: 192.168.2.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults       13) Update from console
5) Reboot system                  14) Enable Secure Shell (sshd)
6) Halt system                   15) Restore recent configuration
7) Ping host                     16) Restart PHP-FPM
8) Shell

Enter an option:
Message from syslogd@PF101 at May 13 10:23:12 ...
php-fpm[991821]: /index.php: Successful login for user 'admin' from: 192.168.2.201 (Local Database)
```

Site B :PF101VPN

- WAN : 192.168.101.2 (~~Site Simu :192.168.50.2~~)
- LAN : 192.168.3.1 (~~site simu :192.168.101.3.1~~)
- Réseau : 192.168.3.0/24

```
PING 192.168.2.1 (192.168.2.1): 56 data bytes
--- 192.168.2.1 ping statistics ---
3 packets transmitted, 0 packets received, 100.0% packet loss

Press ENTER to continue.
QEMU Guest - Netgate Device ID: a80a14ae91b491fe4b1d

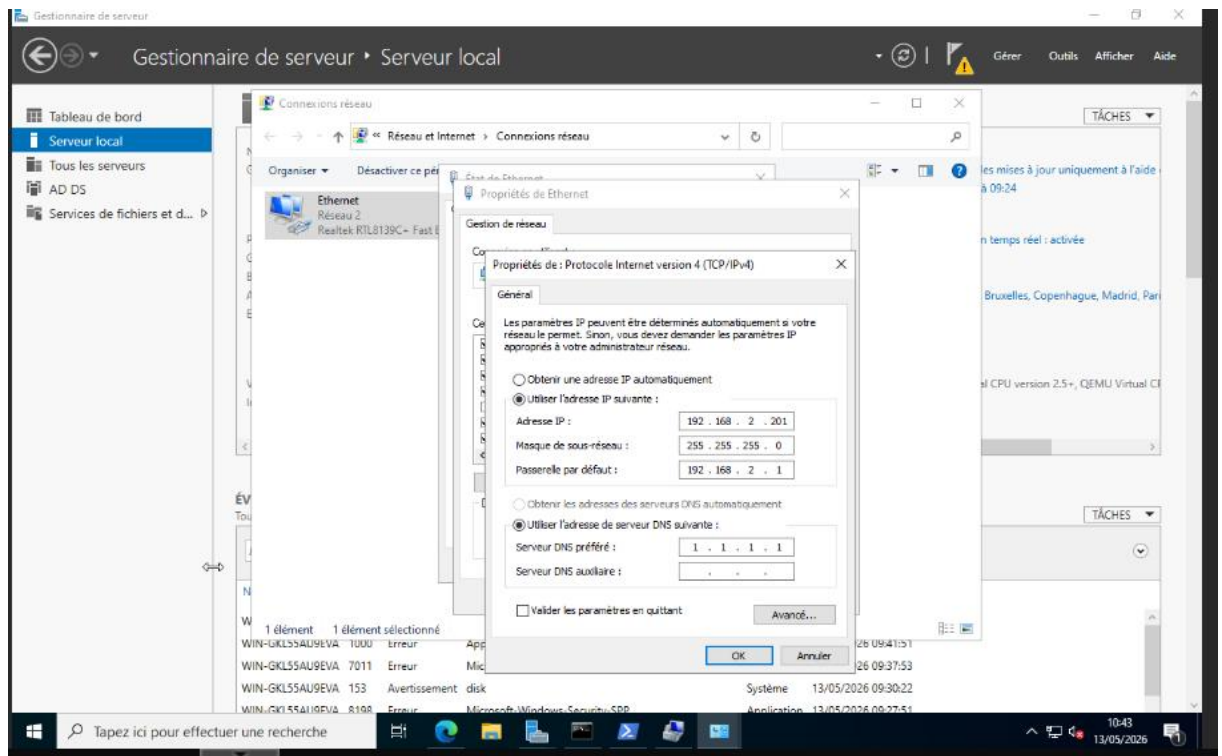
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> vtnet0 -> v4: 192.168.101.2/24
LAN (lan) -> vtnet1 -> v4: 192.168.3.1/24

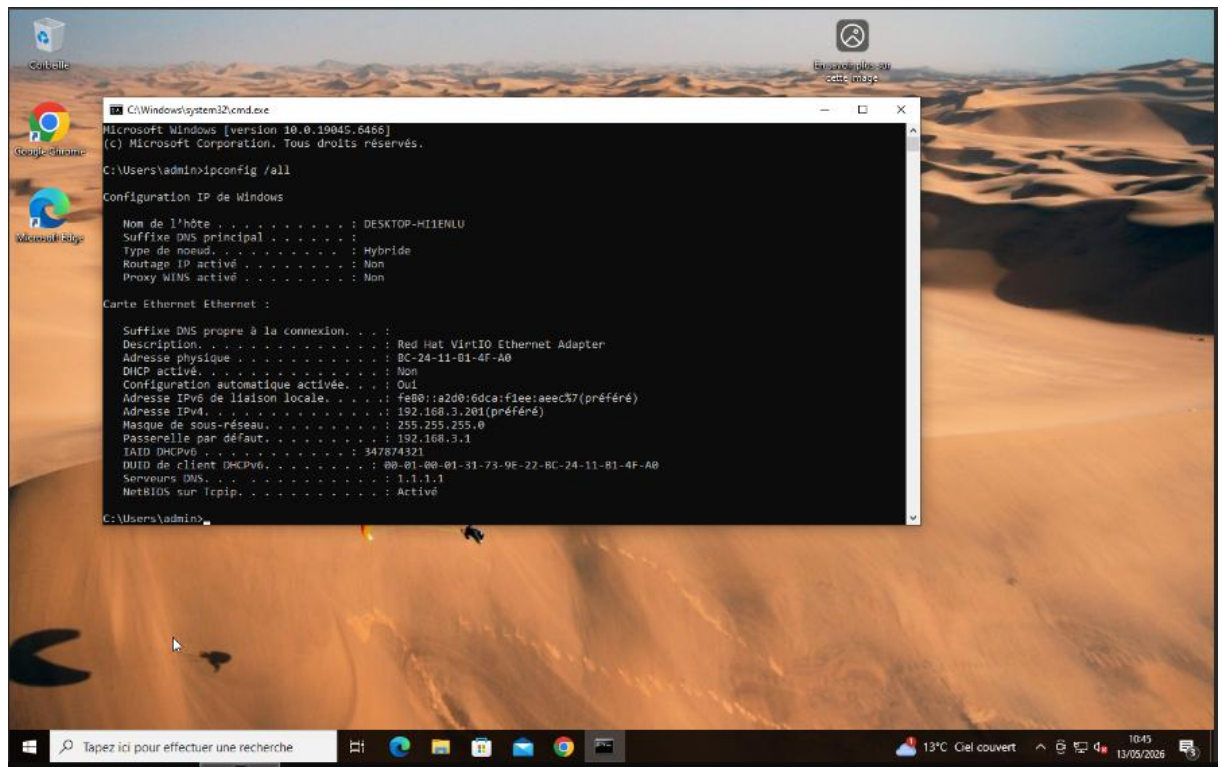
0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults         13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Client A PRODSRV:



Client B Win10VPN :



ÉTAPE 3 — CONFIGURATION IPSEC (PHASE 1)

La phase 1 permet d'établir un tunnel sécurisé entre les deux pare-feux en utilisant IKEv2 et une clé pré-partagée.

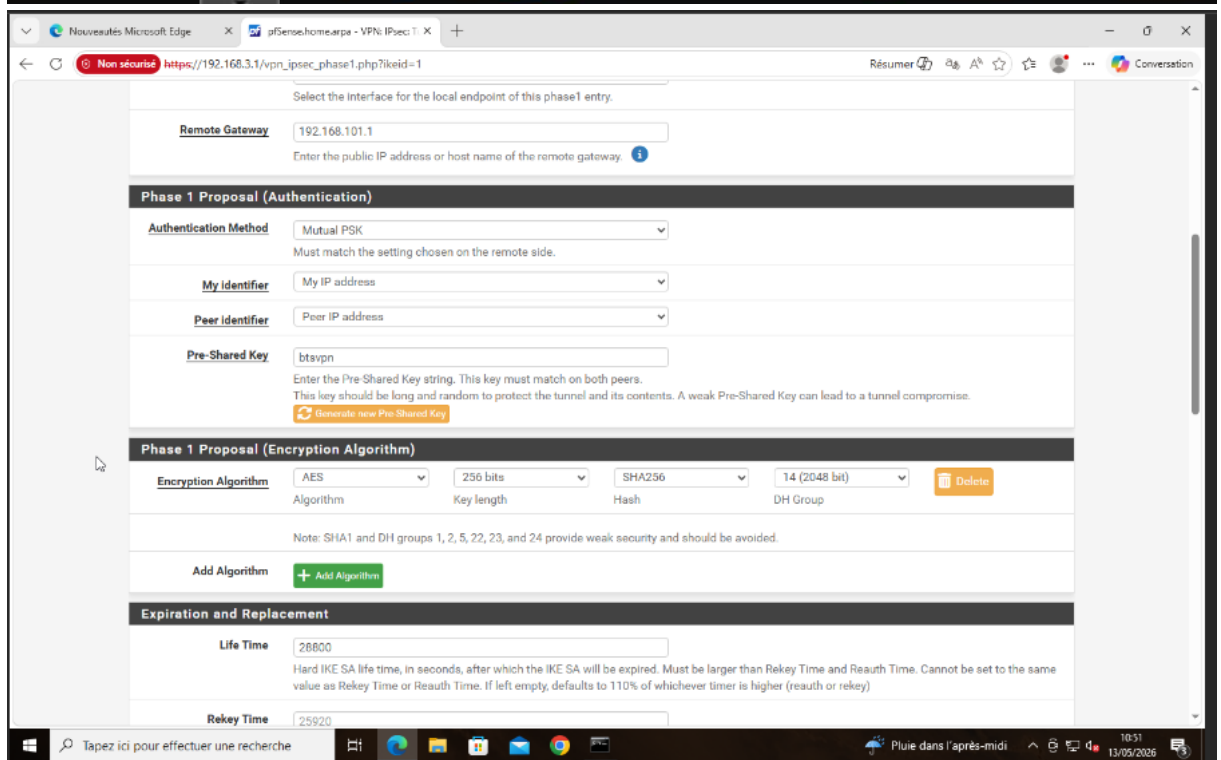
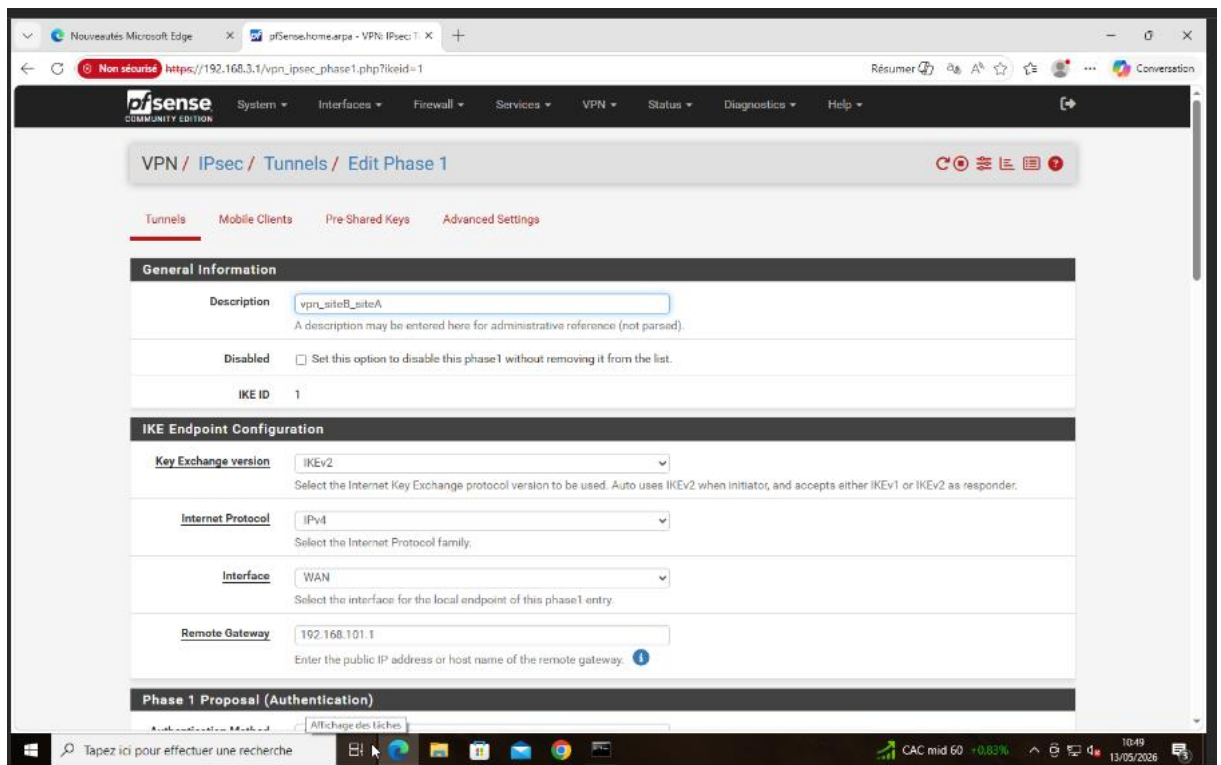
Sur PF101 (Site A)

- Interface : WAN
- Remote Gateway : 192.168.101.2
- Auth : Mutual PSK
- Clé : btspvn
- Chiffrement : AES 256
- Hash : SHA256
- DH Group : 14

Sur pfSense B

Remote Gateway inversé :

- 192.168.101.1



The screenshot shows the pfSense web interface for configuring IPsec Phase 1. The browser address bar indicates the URL is https://192.168.3.1/vpn_ipsec_phase1.php?ikeid=1. The page content includes several configuration sections:

- MOBIKE:** A dropdown menu is set to "Disable". Below it, a note states: "Set this option to control the use of MOBIKE."
- Gateway duplicates:** An unchecked checkbox. The description reads: "Enable this to allow multiple phase 1 configurations with the same endpoint. When enabled, pfSense does not manage routing to the remote gateway and traffic will follow the default route without regard for the chosen interface. Static routes can override this behavior."
- Split connections:** An unchecked checkbox. The description reads: "Enable this to split connection entries with multiple phase 2 configurations. Required for remote endpoints that support only a single traffic selector per child SA."
- PRF Selection:** An unchecked checkbox. The description reads: "Enable manual Pseudo-Random Function (PRF) selection. Manual PRF selection is typically not required, but can be useful in combination with AEAD Encryption Algorithms such as AES-GCM."
- Custom IKE/NAT-T Ports:** Two input fields. The first is labeled "Remote IKE Port" and the second is labeled "Remote NAT-T Port". Below the first field, it says: "UDP port for IKE on the remote gateway. Leave empty for default automatic behavior (500/4500).". Below the second field, it says: "UDP port for NAT-T on the remote gateway".
- Dead Peer Detection:** A checked checkbox labeled "Enable DPD". The description reads: "Check the liveness of a peer by using IKEv2 INFORMATIONAL exchanges or IKEv1 R_U_THERE messages. Active DPD checking is only enforced if no IKE or ESP/AH packet has been received for the configured DPD delay."
- Delay:** An input field containing the value "10". The description reads: "Delay between sending peer acknowledgement messages. In IKEv2, a value of 0 sends no additional messages and only standard messages (such as those to rekey) are used to detect dead peers."
- Max failures:** An input field containing the value "5". The description reads: "Number of consecutive failures allowed before disconnecting. This only applies to IKEv1; in IKEv2 the retransmission timeout is used instead."

At the bottom of the configuration area, there is a blue "Save" button. The footer of the page states: "pfSense is developed and maintained by Netgate. © ESF 2004 - 2020 View license."

ÉTAPE 4 — CONFIGURATION PHASE 2

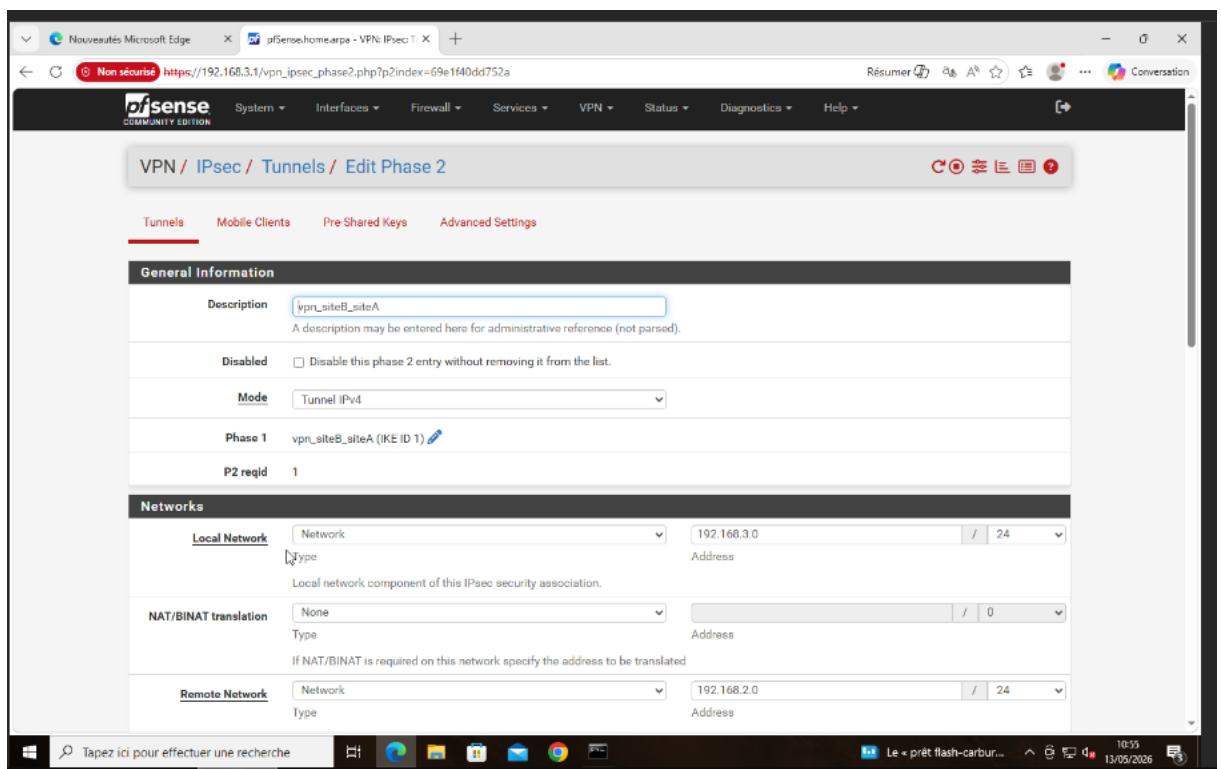
La phase 2 permet de définir les réseaux qui vont communiquer à travers le tunnel VPN.

Site A PF101 :

- Local : 192.168.2.0/24
- Remote : 192.168.3.0/24

Site B PFVPN101

- Local : 192.168.3.0/24
- Remote : 192.168.2.0/24



Non sécurisé https://192.168.3.1/vpn_ipsec_phase2.php?p2index=69e1f40dd752a

Remote Network /
Type Address
Remote network component of this IPsec security association.

Phase 2 Proposal (SA/Key Exchange)

Protocol
Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.

Encryption Algorithms ☒ AES
☐ AES128-GCM
☐ AES192-GCM
☐ AES256-GCM
☐ CHACHA20-POLY1305

Hash Algorithms ☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC
Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group
Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time
Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3600.

Rekey Time
Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Microsoft Suite. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that

Non sécurisé https://192.168.3.1/vpn_ipsec_phase2.php?p2index=69e1f40dd752a

Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group
Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time
Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3600.

Rekey Time
Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time
A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host
Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive ☐ Enable periodic keep alive check
Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.

[Save](#)

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ÉTAPE 5 — CONFIGURATION FIREWALL :

Des règles firewall sont nécessaires pour autoriser le trafic VPN.

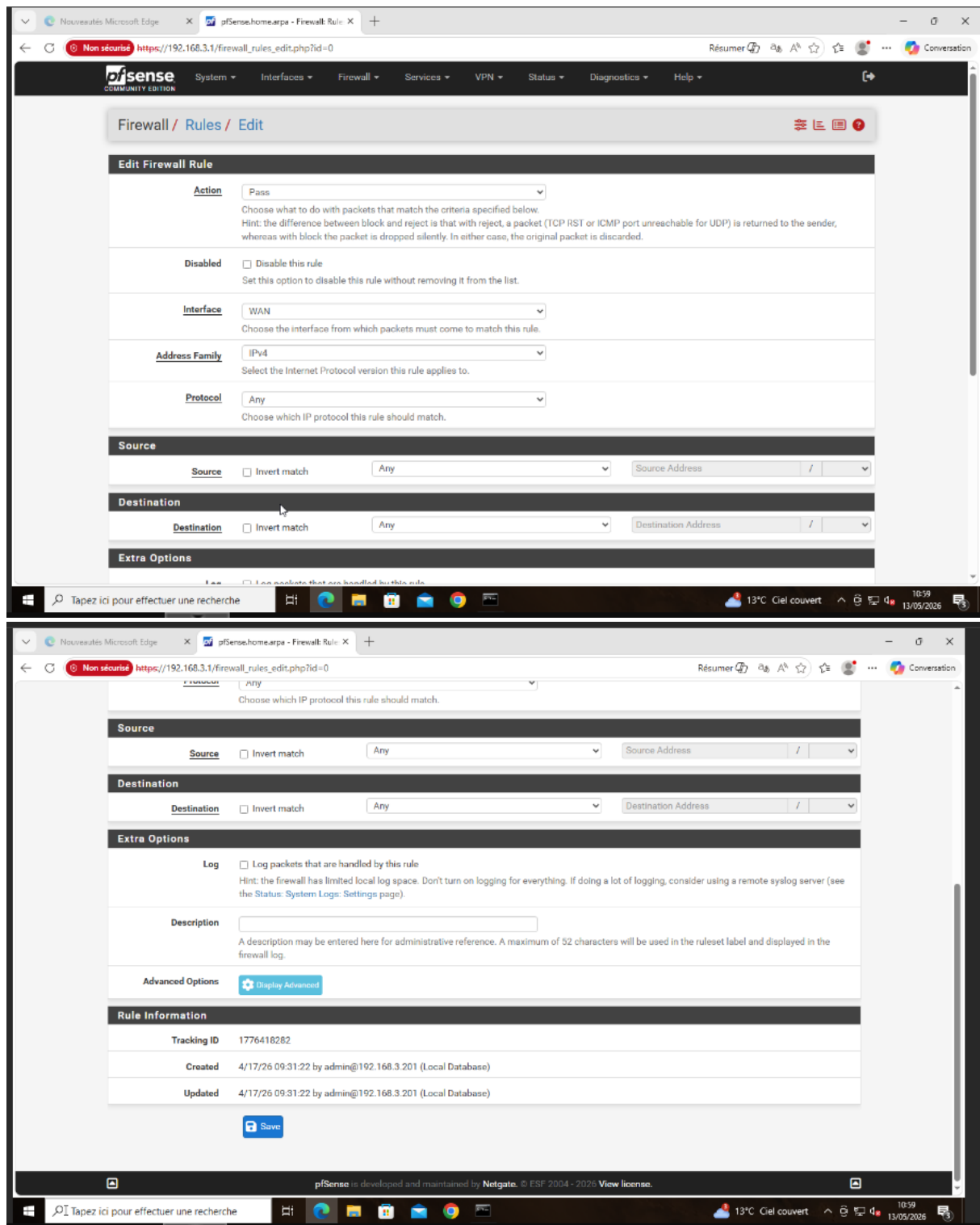
Dans chaque pfSense :

Interface IPsec :

- Autoriser ANY → ANY

Interface LAN :

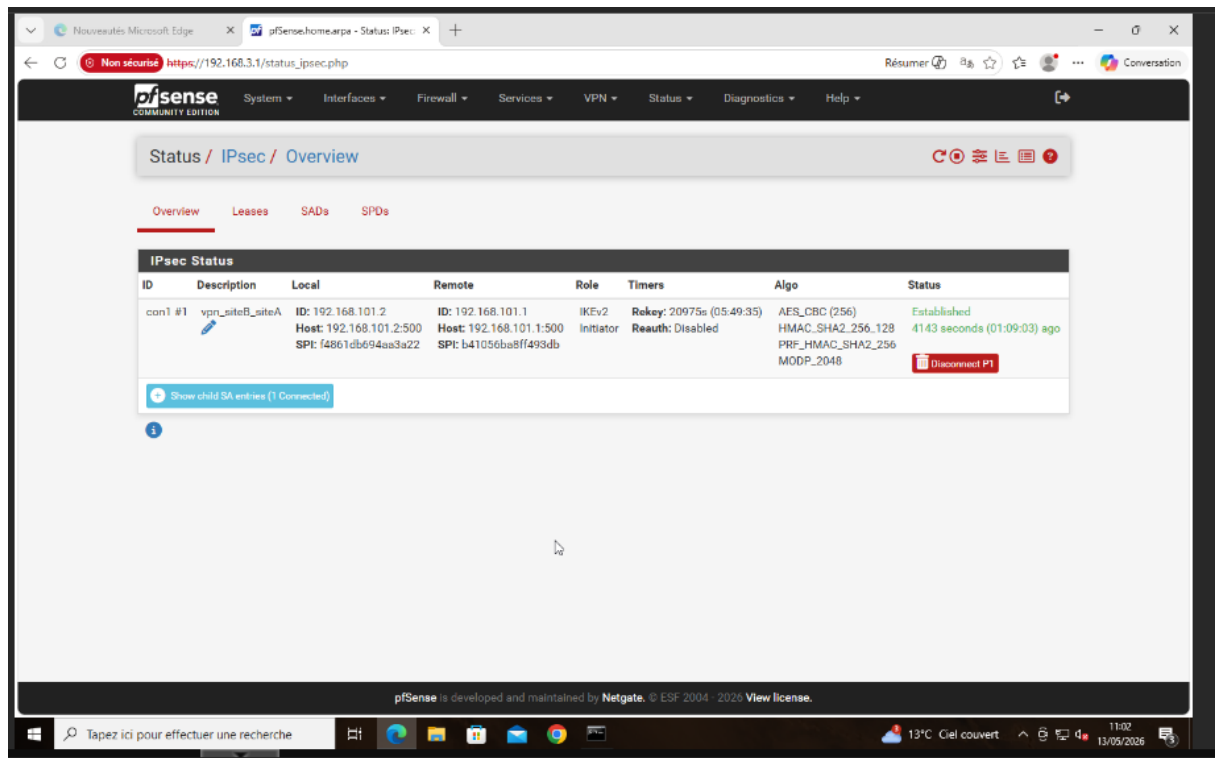
- Autoriser vers réseau distant



- Règles IPsec
- Règles LAN

ÉTAPE 6 — ÉTABLISSEMENT DU TUNNEL

Une fois la configuration terminée, le tunnel IPsec est établi entre les deux sites.



- Status → IPsec (Connected)

ÉTAPE 7 — TESTS DE CONNECTIVITÉ

Des tests de communication ont été réalisés pour valider le fonctionnement du VPN.

Tests :

Depuis Site A :

ping 192.168.3.201

Depuis Site B :

ping 192.168.2.1

ping 192.168.101.1 sur le pf

```

1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults 13) Update from console
5) Reboot system             14) Enable Secure Shell (sshd)
6) Halt system               15) Restore recent configuration
7) Ping host                 16) Restart PHP-FPM
8) Shell

Enter an option: 7

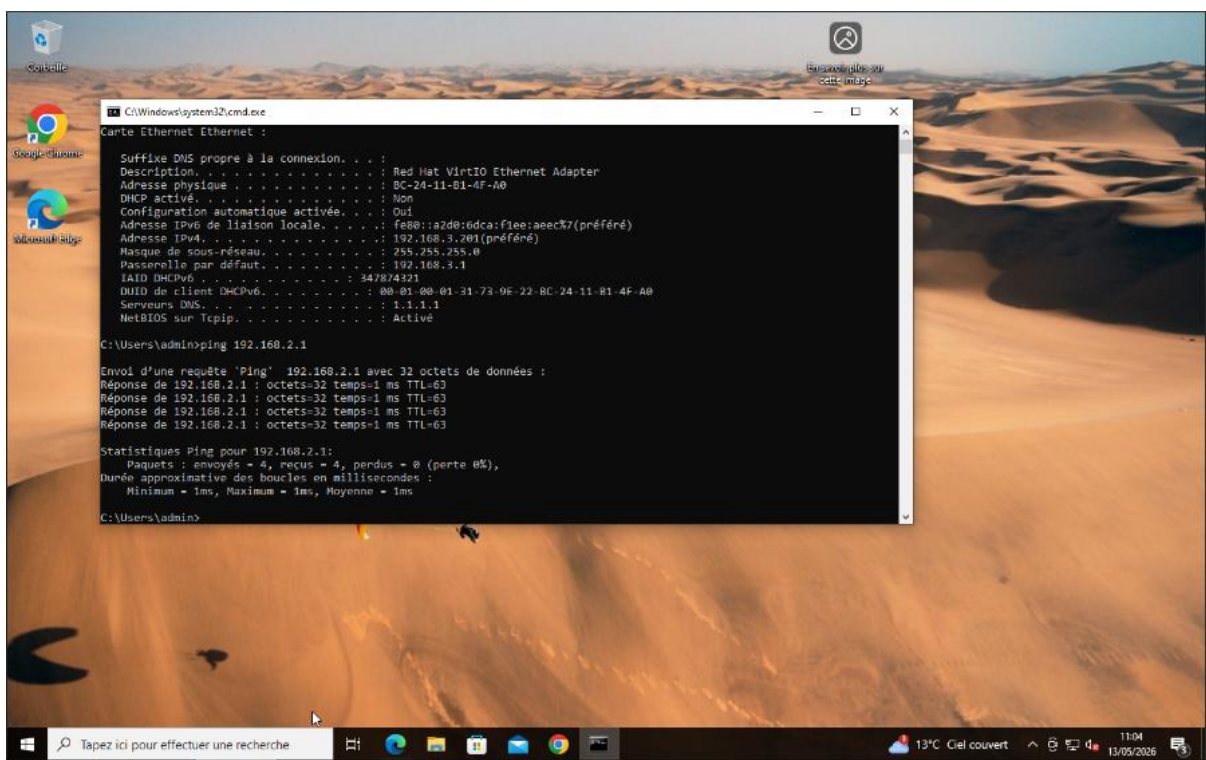
Enter a host name or IP address: 192.168.101.1

PING 192.168.101.1 (192.168.101.1): 56 data bytes
64 bytes from 192.168.101.1: icmp_seq=0 ttl=64 time=0.633 ms
64 bytes from 192.168.101.1: icmp_seq=1 ttl=64 time=0.556 ms
64 bytes from 192.168.101.1: icmp_seq=2 ttl=64 time=0.565 ms

--- 192.168.101.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.556/0.585/0.633/0.034 ms

Press ENTER to continue.

```



- Ping réussi
- CMD ou terminal

ÉTAPE 8 — RÉSULTAT

Le VPN site-to-site fonctionne correctement. Les deux réseaux peuvent communiquer de manière sécurisée.